NOHELIA ESTEFHANIA TACCA APAZA

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RESEARCH INTERESTS: Indigenous Language Revitalization (Southern Quechua) • Low-Resource Natural Language Processing (NLP) & Computer-Assisted Language Learning (CALL) • Embodied Cognition & Second Language Acquisition (SLA) • Andean Digital Humanities & Data Sovereignty

EDUCATION

2025 – 2026 Visiting International Student & Audit Student (Non-Degree)
Cornell University, Ithaca, NY, USA - *Sponsored by the Fulbright Program*

- Completed graduate and undergraduate coursework in Latin American Studies and Pedagogical Methodologies while serving as primary Quechua instructor.
- **Graded Coursework:** International Teaching Assistant Development Program Course I (Fall 2025); International Teaching Assistant Development Program Course II (Spring 2026).
- **Audited Coursework (Non-Degree):** Indigenous Issues in Global Perspectives; Exploring Strategies for Teaching and Learning; Intermediate Portuguese for Spanish Speakers I; The First American University.

2020 – 2024 Bachelor of Science (B.S.) in Systems Engineering
Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

- Research focus: Software architecture, human-centered computing, and educational technologies for indigenous language preservation and social impact.

2018 (Aug – Dec) International Exchange Student, International Business & Research Seminar Participant
Universidad Santo Tomás (USTAMED), Medellín, Colombia

2017 (Mar – Jul) International Exchange Student, Public Accounting Program (CRISCOS Scholar)
Universidad Católica del Norte (UCN), Antofagasta, Chile

2014 – 2018 Bachelor of Science (B.S.) in Accounting
Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

- Quantitative training in financial auditing, resource management, and socio-economic project evaluation.

EXPERIENCE

2025 – 2026 Quechua Language Instructor (Fulbright FLTA Fellow)
Latin American and Caribbean Studies Program / Department of Romance Studies, Cornell University, Ithaca, NY, USA (Fall 2025 – Spring 2026)

- **Instructed Courses:**
    - Fall 2025 – Elementary Quechua I;
    - Spring 2026 – Elementary Quechua I and Elementary Quechua II.
- Facilitated university-level Quechua language instruction for undergraduate and graduate students, designing original syllabi, digital multimedia resources, and interactive morphosyntactic exercises.
- Spearheaded cultural immersion workshops, including Andean dance and linguistic breakdown sessions (*Kuntur Tusuy*), strengthening heritage and second-language learner engagement across campus.
- Facilitated weekly *Quechua Conversation Hours* at the Cornell University Language Resource Center (Fall 2025 – Spring 2026).

2025 – Present Quechua Language Instructor
Language Center, Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru (Feb 2025 – Jul 2025; Sep 2025 – Present)

- Instruct undergraduate, graduate, and community students in Southern Quechua grammar, oral tradition, and Andean epistemology, integrating digital tools to bridge urban academic spaces with indigenous linguistic heritage.

2024 – Present Founder & Lead Instructor - "AI for Children" (*IA para Niños*) & Educational Robotics Workshops
Nuñoa District (Melgar, Puno, Peru) - In collaboration with HOOP Peru & National Central University (NCU, Taiwan)

- Spearheaded the 1st, 2nd, and 3rd editions of the Educational Robotics Workshop and led the *"AI for Children"* initiative in the native Quechua-speaking community of Nuñoa, Puno.
- Designed culturally responsive curricula introducing rural Andean children to Artificial Intelligence, algorithmic logic, and robotics in partnership with international volunteers from Taiwan National Central University, Colombia, the United States, and Peru.

2024 – Present Founder & Lead Instructor - "Discovering Opportunities" (*Descubriendo Oportunidades*)
Nuñoa District (Melgar, Puno, Peru) - In collaboration with National Central University (NCU, Taiwan)

- Organized and led a mentorship speaker series connecting high school students in rural Puno with higher education scholarships and post-secondary pathways, featuring role models from similar socioeconomic and cultural backgrounds.

2024 – 2025 Software Development Pre-Professional Intern
Globant University, Peru (Aug 2024 – Apr 2025)

- Completed software engineering training and collaborative agile development projects within a leading global digital transformation organization.

2022 – 2023 Head Coach in Coding & Robotics Program
Helping Overcome Obstacles in Peru (HOOP Peru), Arequipa, Peru (May 2022 – Dec 2023)

- Directed coding (Scratch) and educational robotics instruction for underserved children in Arequipa, mentoring volunteer instructors and developing project-based STEM learning modules.

RESEARCH & DIGITAL HUMANITIES PROJECTS

2025 – Present Lead Researcher - *Embodied Cognition & Andean Choreography in Quechua SLA ("Kuntur Tusuyta Tususunchis")*

- Co-developed an interdisciplinary pedagogical framework repositioning traditional Andean dance not merely as folkloric performance, but as a measurable cognitive technology for Quechua Second Language Acquisition (SLA).
- Integrated Embodied Cognition and Total Physical Response (TPR).
- Designed an assessment roadmap combining Pre/Post-test SLA retention matrices with NLP-driven phonetic accuracy and response latency tracking.

2025 – Present Co-Investigator - *Paqocha: A Retrieval-Augmented Model for Spanish to Cusco-Collao Quechua Translation*

- Architected an innovative Spanish-to-Quechua (Cusco-Collao variety) machine translation system built on a Retrieval-Augmented Generation (RAG) framework to address morphosyntactic complexity in low-resource agglutinative languages.

2025 – Present Co-Investigator - *Paqocha Analyzer: Audio Concordance & Morphological Breakdown Platform for Puno Quechua*

- Designed and deployed a reproducible, low-cost YouGlish-style Computer-Assisted Language Learning (CALL) web platform leveraging the spontaneous-speech corpus of Common Voice Puno Quechua (qxp).
- Built an automated data ingestion pipeline filtering 1,222 transcribed audio recordings and processing them through a deterministic, rule-based morphological analyzer that decomposes agglutinative words into roots and suffixes via iterative right-to-left stripping alongside word-level forced timestamp alignment.
- An interactive web interface with word search, synchronized audio playback, and morphosyntactic visualization. https://paqocha-analyzer.vercel.app/

2022 – Present Assistant Researcher
Center for Research, Technology Transfer, and Software Development (CiTeSoft), UNSA, Arequipa, Peru (Aug 2022 – Present)

- Conduct multidisciplinary research at the intersection of Systems Engineering, computational linguistics, and human-centered software development to address educational and linguistic inequities in southern Peru.

2023 – 2024 Quechua Linguistic Data Specialist & Transcriber (Freelance)
Sigma AI, Remote

- Transcribed, segmented, and annotated spontaneous oral corpora in Puno (Collao) Quechua (qxp).

CONFERENCE PRESENTATIONS & PROCEEDINGS

2026 **Tacca Apaza, N. E.**, & Machaca Condori, E. E. J. (2026). *"Dance as a Vehicle for the Preservation and Revitalization of the Quechua Language: The Case of the Workshop 'Kuntur Tusuyta Tususunchis'."*

- **Conference on Indigenous Languages of Latin America (CILLA) 2026**, Department of Linguistics & LLILAS Benson Latin American Studies and Collections, *The University of Texas at Austin*, Austin, TX. *Accepted and presented.*
- **Symposium on American Indian Languages (SAIL) 2026**, *The University of Arizona*, Tucson, AZ. *Accepted and presented.*
- **First Glocal Conference on Quechua Languages and Cultures 2026**, Quechua Language and Culture Program at *University of Pennsylvania* & Hamilton Lugar School of Global and International Studies at *Indiana University Bloomington*, Philadelphia, PA. *Accepted and presented.*

2026 Merma Portocarrero, J. F., **Tacca Apaza, N. E.**, Fernandez, R., & Luque Ccosi, P. A. (2026). *"Paqocha Analyzer: An Audio Concordancer and Deterministic Morphological Analyzer for Puno Quechua."*

- **International Conference on Information Management and Big Data (SIMBig) - Shared Task**, *Universidad Católica San Pablo*, Arequipa, Peru. *Accepted, presented, and awarded Honorable Mention.*

2025 – 2026 Merma Portocarrero, J. F., & **Tacca Apaza, N. E.** (2025–2026). *"Paqocha: A Retrieval-Augmented Model for Spanish to Cusco Quechua Translation."*

- **First Glocal Conference on Quechua Languages and Cultures 2026**, *University of Pennsylvania* & *Indiana University Bloomington*, Philadelphia, PA. *Accepted and presented.*
- **Symposium on American Indian Languages (SAIL) 2026**, *The University of Arizona*, Tucson, AZ. *Accepted and presented.*
- **Society for the Study of the Indigenous Languages of the Americas (SSILA) Annual Meeting 2026**, New Orleans, LA. *Accepted and presented.*

INVITED TALKS, CULTURAL WORKSHOPS & ACADEMIC EVENTS

2026 (Apr 9) **Invited Workshop Leader:** *"An Immersive and Interactive Exploration of Quechua Language and the Living Traditions of the Andes"*, SUNY Oneonta, Oneonta, NY, USA.

2026 Invited Guest Speaker & Workshop Co-Leader: *"The Living Voice of the Andes: An Interactive Journey into Quechua Culture with Nohelia Tacca and Eddy Machaca"*, Mount Holyoke College, South Hadley, MA, USA.

2026 Invited Guest Speaker: Quechua Language and Culture Program (Invited by Prof. Odi Gonzales), New York University (NYU), New York, NY, USA.

2025 (Nov 11) Invited Panelist: *"Fulbright Brings the World to Us"*, SUNY Oneonta (Morris Hall), Oneonta, NY, USA.

2025 – 2026 Workshop Leader & Guest Presenter, Cornell University, Ithaca, NY, USA:

- *Language Resource Center, World Language Day Presentations (May 2025 & May 2026):* "The Secret of the Andes: Where Do Llamas and Alpacas Come From?"
- *Department of Romance Studies, Guest Lectures in Spanish for Heritage Speakers (Invited by Prof. Mary K. Redmond):*
    - To be and speak Quechua - "Ser y hablar Quechua" (Fall 2025, two groups)
    - Beyond Our Roots - "Llaqtanchismanta, más allá de nuestras raíces" (Spring 2025, two groups, with Prof. Eddy Machaca);
    - Where We Come From, Where We Are Going - "De dónde venimos, hacia dónde vamos" (with Prof. Eddy Machaca).
- *Cornell Quechua Program Cultural Workshops (2025–2026):*
    - Leader of
        - Andean Dances – "Tususunchis",
        - Harvesting Andean Potatoes – "Papa Hasp'iy",
        - Andean Textiles – "Punchunchista Sirasunchis", and
        - Making Our Ponchos – "Punchu Ruway";
    - Co-leader of
        - Learning About Andean Music Through the Pan Flute – "Takisunchis Antarawan".

FELLOWSHIPS, GRANTS & ACADEMIC HONORS

- **2026 - Aspire Social Impact Seed Fund Award**, Aspire Institute - Awarded funding to develop the Leadership & Community Empowerment Initiative entirely in the Quechua language in Santa Rosa (Melgar, Puno, Peru).
- **2026 - SIMBig Shared Task Honorable Mention**, Universidad Católica San Pablo, Arequipa, Peru.
- **2025 – 2026 - Fulbright Foreign Language Teaching Assistant (FLTA) Fellowship**, U.S. Department of State & Institute of International Education (IIE) - Hosted at Cornell University, USA.
- **2025 – 2026 - Language Resource Center (LRC) Grant**, Cornell University - Project Director for *"Pachamamawan kawsayninchismanta, Our Life with the Pachamama: Real-Life Audiovisual Resources for Elementary Quechua"*.

- **2025 – 2026 - Language Resource Center (LRC) Grant**, Cornell University - Project Collaborator with Prof. Vanessa Gubbins on *"Intiq wawankunaq willakuyninkuna: Andean Stories in Elementary Quechua"* (Storybook Flipbook).
- **2025 - Aspire Leaders Program Fellow (Cohort 3)**, Aspire Institute.
- 2024-2025 - Code Your Future Scholarship for learning Backend Developer with Java
- **2023 - International Student Festival in Trondheim (ISFiT) Selected Delegate**, Trondheim, Norway - Represented Peru in global dialogues on education, equity, and social impact.
- **2022 – 2023 - College Horizons Outreach Program (CHOP+) Fellowship**, U.S. Embassy in Peru (Jan 2022 – Oct 2023) - Competitive scholarship for indigenous Quechua youth focused on advanced English, social entrepreneurship, and community leadership.
- **2022 - Grace Hopper Celebration (GHC) Scholar**, AnitaB Organization - Awarded competitive fellowship recognizing emerging women leaders in computing and technology.
- **2017 - CRISCOS Academic Mobility Scholarship**, Consejo de Rectores por la Integración de la Subregión Centro Oeste de Sudamérica - Full academic exchange grant at Universidad Católica del Norte, Chile.
- **2014 – 2018 - Beca Permanencia (Undergraduate Academic Excellence & Retention Scholarship)**, PRONABEC – Peruvian Ministry of Education.

VOLUNTEER

2026 Project Lead - *Aspire* Youth Leadership & Community Empowerment Initiative
Santa Rosa (Melgar, Puno, Peru)

- Led educational empowerment, leadership mentoring, and digital inclusion workshops conducted in Quechua for high-Andean youth in Santa Rosa, Puno, fostering pathways to higher education and community-driven innovation.

2024 – Present Global Shaper - World Economic Forum (Arequipa Hub)
Arequipa, Peru (May 2024 – Present)

- Co-design and execute high-impact community projects focused on educational equity (SDG 4), youth leadership, and decentralized technological access in southern Peru.

2024 – 2025 Organizer - Workshop *"Empowering Andean Youth Through Social Leadership"*
Melgar and Azángaro Provinces, Puno, Peru

- Delivered youth leadership workshops at Domingo Savio High School (Nuñoa), Túpac Amaru High School (Nuñoa), Siervos de Dios High School (Crucero), and San Agustín High School (Samán).

2017 – 2024 Volunteer

- **Santa Casa de Diadema / AIESEC Global Volunteer** (São Paulo, Brazil | Jan 2022 – Feb 2022): Taught Spanish language and Latin American cultural workshops to children aged 11–13.

- **Fraternidad Cultural Pasiones** (Arequipa, Peru | May 2024 - Dec 2024) Promoted the preservation, and performance of traditional Peruvian and Andean choreographies.
- **Kallpachakusun,** (Peru | Dec 2020 - Mar 2021). Learning, cognitive skill development, and personal and professional growth and research about social problems.
- **Proyecto Pasaporte** (Peru, Bolivia, and Paraguay | Dec 2020 – Aug 2021): Delivered virtual mathematics for high school students.
- **Ayni Educativo** (Peru | 2021): Delivered virtual early childhood English for children across the Andean and Southern Cone regions during the COVID-19 pandemic.
- **TECHO** (Antofagasta, Chile | May 2017 – Jul 2017): Provided academic mentoring and community integration programming in underserved settlements.

LANGUAGES

- **Languages:**
    - Quechua (Native Speaker),
    - Spanish (Native Speaker),
    - English (Advanced / Fluent, C1),
    - Portuguese (Elementary Proficiency, A2),
    - French (Basic Proficiency, A1).